21 Effect size

Subject content	Additional information
21.1 Know the notion of effect size as a complementary methodology to standard significance testing, and apply in authentic contexts.	Students should be aware that interpretation of effect size depends on the context involved and that effect size simply quantifies the size of the difference between two means (rather than its statistical significance). Students should also be aware that a p-value, in the context of testing for statistical significance, depends on both the effect size and the sample size.
21.2 Know and use Cohen's d in simple situations.	Students should be aware of the standard guideline boundaries for interpreting the value of Cohen's d. $0.2 \leqslant d < 0.5$ small effect size $0.5 \leqslant d < 0.8$ medium effect size $0.8 \leqslant d$ large effect size

Statistical Enquiry Cycle (SEC)

The Statistical Enquiry Cycle (SEC) underpins the study of Statistics. Students need to be able to apply the knowledge and techniques outlined in this section within the framework of the SEC. The cycle covers five stages:

- initial planning
- data collection
- data processing and presentation
- interpretation of results
- evaluation and review.

The detail of the SEC is provided below. During their learning students should develop their understanding of the SEC through a variety of authentic contexts. Practical experience of the cycle is integral to their understanding of the principles of the SEC.

Students should anticipate that they may occasionally be presented, within the context of applying elements of the SEC cycle, unfamiliar acronyms or scenarios in real-life data.

A Initial planning	
Students must understand the importance of initial planning when designing a line of enquiry or investigation including:	
1	identifying factors that may be related to the problem under investigation
2	defining a question or hypothesis (or hypotheses) to investigate
3	deciding what data to collect, and how to collect and record it, giving reasons
4	engaging in exploratory data analysis in order to investigate the situation
5	developing a strategy for how to process and represent the data giving reasons
6	justifying the proposed plan with regards to ensuring a lack of bias.

B Data collection	
Students must recognise the constraints involved in sourcing data including:	
1	when designing unbiased collection methods for primary sample data
2	when researching sources of secondary data, including from reference publications, the internet and the media
3	the importance of declaring the data collection methodology, including appreciating the importance of acknowledging sources
4	appreciating the inherent bias that may be incorporated through the use of leading questions either by accident or through agenda driven design.

C Data processing and presentation	
Students must understand a range of techniques in order to process, represent and discuss data including:	
1	organising and processing data, including an understanding of how technology can be used
2	make inferences about the population using appropriately chosen diagrams and summary measures to represent data including an understanding of outputs generated by appropriate technology
3	appreciating how to avoid misrepresentation of data.

D Interpretation of results	
Students must appreciate the need to consider the context of the problem when interpreting results:	
1	analysing/interpreting diagrams and calculations/measures
2	drawing together conclusions that relate to the questions and hypotheses addressed
3	using appropriate tests to determine the statistical significance of the findings
4	discussing the reliability of findings
5	Students must show an understanding of the importance of the clear and concise communication of findings and key ideas, and awareness of target audience.

E Evaluation and review	
Students must be able to understand the importance of evaluating statistical work including:	
1	identifying weaknesses in approaches used to collect or display data
2	recognising the limitations of findings by considering sample size and sampling technique
3	suggesting improvements to statistical processes or presentation
4	refining processes to elicit further clarification of the initial hypothesis.

3 Assessment information

Paper 1: Data and Probability **(Paper code: 9ST0/01)**
• First assessment: May/June 2019. • The assessment is 2 hours. • The assessment is out of 80 marks. • Students must answer all questions. • Calculators can be used in the examination. • The booklet 'Statistical Formulae and Tables' will be provided for use in the assessments.
Content assessed Questions may be set on any of the following topics: 1 – Numerical measures, graphs and diagrams 2 – Probability 3 – Population and samples 4 – Introduction to probability distributions 5 – Binomial distribution 6 – Normal distribution 7 – Correlation and linear regression (except 7.2) 11 – Bayes' theorem 12 – Probability distributions 13 – Experimental design 18 – Exponential and Poisson distributions Statistical Enquiry Cycle (SEC)